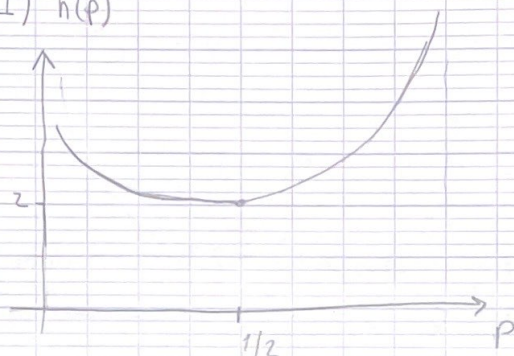


Elias Masouli

## Detection Theory - Homework 2

1)  $h(p)$



$$2) e^{\lambda x} = e^{\lambda x \left( \frac{b-a}{b-a} \right)} = e^{\frac{\lambda}{b-a} (bx - ax + ab - ab)}$$

$$= e^{\lambda \left( a \frac{b-x}{b-a} + b \frac{x-a}{b-a} \right)}$$

Since  $\frac{b-x}{b-a} + \frac{x-a}{b-a} = 1$  and  $e^x$  is convex

$$e^{\lambda x} \leq \frac{b-x}{b-a} e^{\lambda a} + \frac{x-a}{b-a} e^{\lambda b}$$

$$\mathbb{E}[e^{\lambda x}] \leq \frac{b - \mathbb{E}[x]}{b-a} e^{\lambda a} + \frac{\mathbb{E}[x] - a}{b-a} e^{\lambda b}$$

3) •  $\mathbb{1}_{x \geq 0} \leq e^{\lambda x}$

For  $x < 0$ :  $0 \leq e^{\lambda x}$  ✓

For  $x = 0$ : we have equality

For  $x > 0$ :  $1 \leq e^{\lambda x} \Leftrightarrow 0 \leq \lambda x$  ✓ for  $\lambda > 0$

$$\bullet \mathbb{1}_{\{S_1 - \mathbb{E}\{S_1\} - t \geq 0\}} \leq e^{\lambda [S_1 - \mathbb{E}\{S_1\} - t]}$$

$$\bullet \mathbb{P}[S_1 \geq (p+t)] = \mathbb{P}[S_1 \geq \mathbb{E}\{S_1\} + t] = \mathbb{P}[S_1 - \mathbb{E}\{S_1\} - t \geq 0]$$

$$= \mathbb{E}[\mathbb{1}_{\{S_1 - \mathbb{E}\{S_1\} - t \geq 0\}}] \leq \mathbb{E}[e^{\lambda [S_1 - \mathbb{E}\{S_1\} - t]}] =$$

$$= \mathbb{E}[e^{-\lambda (p+t)} e^{\lambda S_1}] = e^{-\lambda (p+t)} \mathbb{E}[e^{\lambda S_1}]$$

$$= e^{-\lambda (p+t)} \prod \mathbb{E}[e^{\lambda x_i}]$$

$x_i$  are independent

$$4) \mathbb{E}[e^{\lambda x}] \leq 1 - \mathbb{E}[x] + \mathbb{E}[x] e^{\lambda}$$

$$\text{Set } p_i = \mathbb{E}[X_i]$$

$$\prod_{i=1}^l \mathbb{E}[e^{\lambda x_i}] \leq \prod_{i=1}^l (1 - p_i + p_i e^{\lambda})$$

The inequality from 2) holds with equality when  $a=x$  or  $b=x$

Being  $a=0$  and  $b=1$ , if  $X_i$ 's are Bernoulli variables, they only take the values 0 or 1. Therefore the inequality becomes an equality.

$$5) \left( \prod_{i=1}^l a_i \right)^{1/l} \leq \frac{1}{l} \sum_{i=1}^l a_i \quad \Leftrightarrow \quad (\log \text{ is increasing})$$

$$\sum \frac{1}{l} \log a_i \leq \log \sum \frac{1}{l} a_i$$

$\log$  is concave so this inequality is just Jensen inequality (note the weights  $1/l$  sum 1)  $\square$

$$6) \prod_i \mathbb{E}[e^{\lambda x_i}] \leq \prod_{i=1}^l (1 - p_i + p_i e^{\lambda})$$

$$\leq \left[ \frac{1}{l} \sum (1 - p_i + p_i e^{\lambda}) \right]^l = (1 - p + p e^{\lambda})^l$$

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$$7) \text{ From 3) : } P[S_1 \geq (p+t)1] \leq e^{-\lambda(p+t)1} \prod_{i=1}^1 E[e^{\lambda x_i}]$$

$$\text{From 6) : } \prod E[e^{\lambda x_i}] \leq (1-p + p e^{\lambda})^1$$

Then

$$P[S_1 \geq (p+t)1] \leq \underbrace{e^{-\lambda(p+t)1} (1-p + p e^{\lambda})^1}_{g(\lambda)} \quad \text{for } \lambda > 0$$

The minimal value of  $g(\lambda)$  is attained at  $\lambda^*$  and so does the minimal value of  $\log g(\lambda)$

$$\log g(\lambda) = -\lambda(p+t)1 + 1[\log(1-p + p e^{\lambda})]$$

Taking the derivative respect to  $\lambda$ :

$$\frac{d \log g(\lambda)}{d \lambda} = 1 \left[ -p-t + \frac{1}{1-p+pe^{\lambda}} p e^{\lambda} \right]$$

$\lambda^*$  is such that

$$\frac{p e^{\lambda^*}}{1-p+p e^{\lambda^*}} - p - t = 0$$

$$\text{Setting } \lambda^* = \frac{\log(1-p)(p+t)}{(1-p-t)p}$$

$$\frac{p(1-p)(p+t)}{p(1-p-t)} \cdot \frac{1}{\frac{1-p+p(1-p)(p+t)}{p(1-p-t)}} - p - t = 0$$
$$\frac{(1-p-t)}{(1-p)} - p - t = 0$$

$$p+t - p - t = 0 \quad \square$$

The bound is tightest for  $\lambda^*$ , obtaining the first Hoeffding inequality.

$$\boxed{1-p > t}$$

$$\lambda^* > 0 \Leftrightarrow \frac{(1-p)(p+t)}{p(1-p-t)} > 1$$

$\Leftrightarrow$

$$(1-p)(p+t) > p(1-p-t)$$

$$t - p^2 + p - pt > p - p^2 - pt$$

$$\boxed{t > 0}$$

$$8) G(t, p) = \frac{p+t}{t^2} \log \frac{p+t}{p} + \frac{1-p-t}{t^2} \log \frac{1-p-t}{1-p}$$

$$\bullet \frac{\partial G}{\partial t} = \frac{t^2 - 2t(p+t)}{t^4} \log \frac{p+t}{p} + \frac{p+t}{t^2} \cdot \frac{p}{p+t} \cdot \frac{1}{p}$$

$$+ \log \frac{1-p-t}{1-p} - \frac{t^2 - 2t(1-p-t)}{t^4} + \frac{1-p-t}{t^2} \cdot \frac{1-p}{1-p-t} \cdot \frac{-1}{1-p}$$

$$= \frac{t - 2(p+t)}{t^3} \log \frac{p+t}{p} + \frac{-t - 2(1-p-t)}{t^3} \log \frac{1-p-t}{1-p}$$

$$\Rightarrow t^2 \frac{\partial G}{\partial t} = \frac{t - 2(p+t)}{t} \log \frac{p+t}{p} + \frac{-t - 2(1-p-t)}{t} \log \frac{1-p-t}{1-p}$$

$$= \left[ 1 - 2 \frac{(1-p)}{t} \right] \log \left( 1 - \frac{t}{1-p} \right) - \left[ 1 - 2 \frac{(p+t)}{t} \right] \log \left( 1 - \frac{t}{p+t} \right)$$

$$= H\left(\frac{t}{1-p}\right) - H\left(\frac{t}{p+t}\right)$$

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## Detection Theory - Homework 2

8)

•  $f(x) = \log(1-x)$

$$f'(x) = \frac{-1}{1-x} \rightarrow f'(0) = -1$$

$$f''(x) = \frac{-1}{(1-x)^2} \rightarrow f''(0) = -1$$

$$f'''(x) = - \frac{-2(1-x)(-1)}{(1-x)^4} = \frac{-2}{(1-x)^3} \rightarrow f'''(0) = -2$$

Taylor expansion around 0:

$$f(x) = -x - \frac{x^2}{2} - \frac{x^3}{3} + \dots$$

$$\Rightarrow H(x) = \left(1 - \frac{2}{x}\right) \left(-x - \frac{x^2}{2} - \frac{x^3}{3} + \dots\right)$$

$$H(x) = 2 + x^2 \left(\frac{2}{3} - \frac{1}{2}\right) + x^3 \left(\frac{2}{4} - \frac{1}{3}\right) \dots$$

- All coefficients are positive so its derivative is positive for  $x > 0 \Rightarrow H(x)$  increasing for  $0 < x < 1$   $\square$

- Since  $H$  is increasing and  $\frac{\partial G}{\partial t} = H\left(\frac{t}{1-p}\right) - H\left(\frac{t}{p+t}\right)$

$$\frac{\partial G}{\partial t} > 0 \text{ iff } \frac{t}{1-p} > \frac{t}{p+t} \Leftrightarrow t > 1-2p \quad \square$$

- When  $t = 1-2p \geq 0$ :  $\frac{\partial G}{\partial t} = 0$ .  $\frac{\partial G}{\partial t} > 0$  if  $t > 1-2p$  and

$\frac{\partial G}{\partial t} < 0$  if  $t < 1-2p$ , so in  $t = 1-2p$  there's a minimum. The value is:

$$G(p) = \frac{p + (1-2p)}{(1-2p)^2} \log \frac{p + 1-2p}{p} + \frac{1-p - 1+2p}{(1-2p)^2} \log \frac{1-p - 1+2p}{1-p}$$

$$G(p) = \log \frac{1-p}{p} \cdot \left(\frac{1-2p}{(1-2p)^2}\right) = \frac{1}{1-2p} \log \frac{1-p}{p} \quad \square$$

9)  $G$  is increasing in  $t$  and we assumed that the values of  $T$  are between 0 and  $1-p$

When  $1-2p \leq 0$ , the smallest value will be attained for the smallest admissible  $t$ :

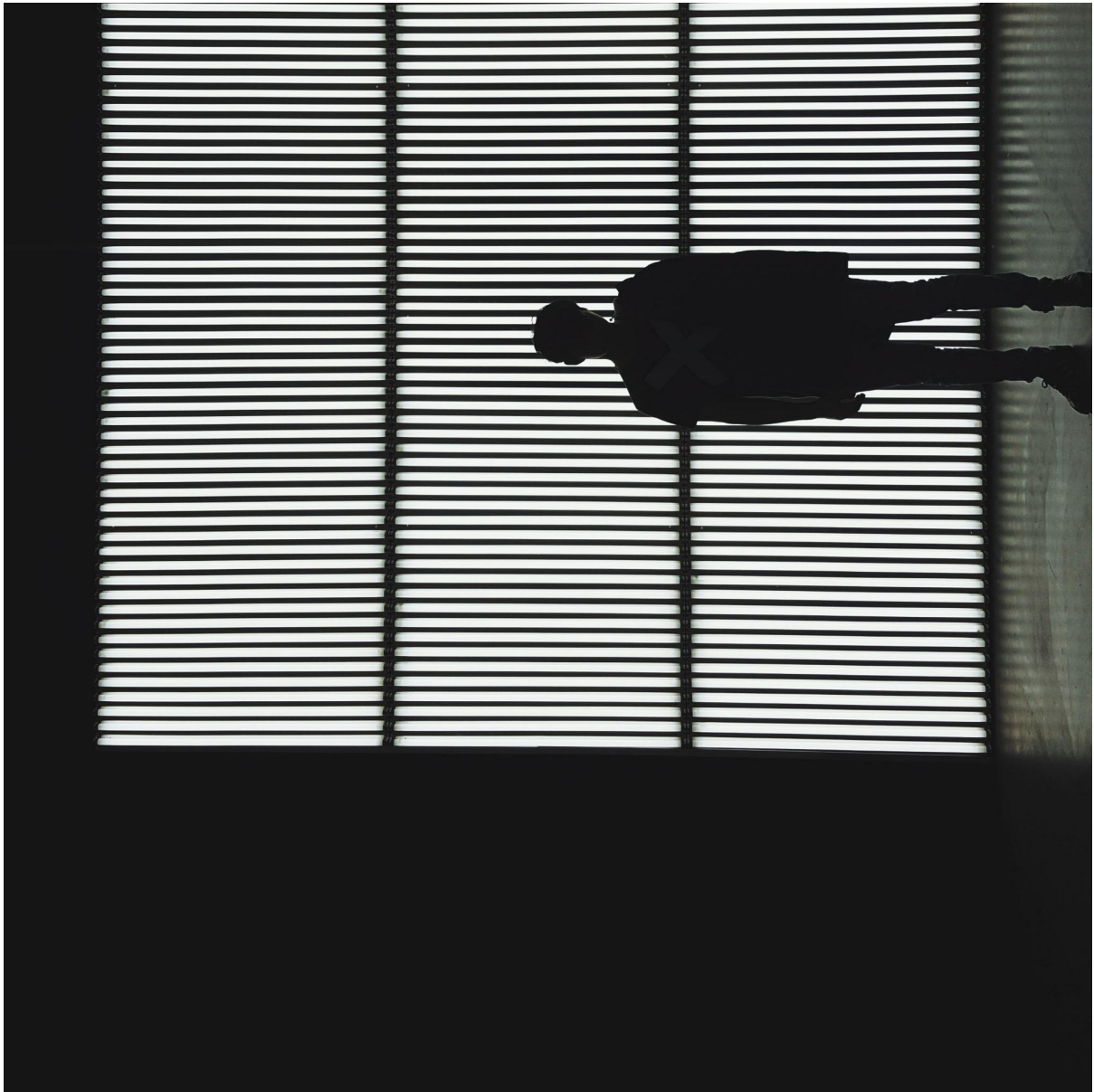
$$\lim_{t \rightarrow 0} G(t, p)$$

$$t \rightarrow 0$$

# Gestalt commentary of an image

Detection Theory - MVA 2021/2022

Elías Masquil



All Gestalt grouping laws manifest in how we perceive the previous image.

### **White square**

By the closure law we perceive a big white square although it has not a defined boundary. If we join the closure law and the continuity of direction, we also see the convexity law for perceiving the square. This white square is also perceived by its color (white) similarity as a single shape.

### **Pattern of thin (horizontal) lines**

Over the white square we perceive a pattern of black lines. We see them in the front as an effect of the T-junctions with the white square. Despite these lines we still see the white square as a single shape because of amodal completion.

In the perception of each line we see the following Gestalt laws: constant width, color similarity, and continuity of direction.

If we move to a larger scale, we identify the set of all parallel lines as a big single pattern. In this, we observe the recursiveness of Gestalt laws. We see parallelism between all lines, vicinity, color and shape similarity.

### **Thick (vertical) lines**

We see two similar vertical lines (constant width, vicinity, similarity of shape and color). They seem to be over the other lines, and again we have T-junctions. Because of the continuity of direction we actually perceive two vertical lines and not small segments split by the horizontal lines. The vertical lines also discover a different pattern at another scale. Instead of having one single white square, we also identify three identical white rectangles. Each of them is defined by closure, convexity, and color consistency. The three of them make a pattern based on the vicinity, continuity of direction (all have the same orientation), and shape and color similarity.

### **Human figure**

Immediately, by past experience, we perceive a human figure. Because of T-junctions, we can see that the subject is in the foreground, while the white square and the lines are in the background. By amodal completion we still see the lines and the square as consistent shapes. We also recognize symmetry on the figure, and see it as a single structure because of the closure law and color similarity as well.

### **Background**

The rest of the image is perceived as background, and as the background has no shape, we don't perceive any other Gestalt.

### **Conflict and past experience**

Note that here there's conflict between Gestalts around the white object. One could perceive this white square with black lines over it or maybe white lines over a black background. The first perception is stronger and those Gestalts prevail in our perception.

The funny thing is that if we look closer, by past experience, we can identify the pattern as a set of fluorescent lamps over a black background. In the end there was indeed a set of white lines over a black background.